HITEC University Taxila

Department of Cyber Security

Course: Essentials of Artificial Intelligence

Lab Manual 5

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Lab Task

Task 1. Write a Python program to find the maximum and minimum value of a given flattened array.

Expected Output:

• Original flattened array:

[[0 1], [2 3]]

• Maximum value of the above flattened array: 3

• Minimum value of the above flattened array: 0

Input

lab5 task 1.py > ...

```python
arr = [[0, 1], [2, 3]]
print("Original flattened array:")
for row in arr:
    print("[", " ".join(map(str, row)), "]")
flattened = [num for row in arr for num in row]
print("Maximum value of the above flattened array:", max(flattened))
print("Minimum value of the above flattened array:", min(flattened))
```

Output

```
task 1.py"
Original flattened array:
[ 0 1 ]
[ 2 3 ]
Maximum value of the above flattened array: 3
Minimum value of the above flattened array: 0
PS S:\AI Lab>
```

Task 2. Create a 3x3 matrix with values ranging from 2 to 10.

Input

lab5 task 2.py > ...

```python
values = list(range(2, 11))
matrix = [values[i:i+3] for i in range(0, len(values), 3)]
print("3x3 matrix with values ranging from 2 to 10:")
for row in matrix:
    print(row)
```

Output

```
PS S:\AI Lab> & C:/Users/HP/AppData/Local/Pro
task 2.py"
3x3 matrix with values ranging from 2 to 10:
[2, 3, 4]
[5, 6, 7]
[8, 9, 10]
PS S:\AI Lab>
```

Task 3. Write a NumPy program to convert an array to a float type. Sample output:

• Original array [1, 2, 3, 4]

• Array converted to a float type: [1. 2. 3. 4.]

Input

```
lab5 task 3.py > ...
import numpy as np
arr = np.array([1, 2, 3, 4])
print("Original array", arr)
float_arr = arr.astype(np.float64)
print("Array converted to a float type:", float_arr)
print("Data type of converted array:", float_arr.dtype)
```

Output

```
PS S:\AI Lab> & C:/Users/HP/AppData/Local/Program
task 3.py"
Original array [1 2 3 4]
Array converted to a float type: [1. 2. 3. 4.]
Data type of converted array: float64
PS S:\AI Lab>
```

Task 4. Multiply 2 x 2 matrix and take transpose of a resultant matrix.

$$M1 = \begin{bmatrix} 5 & 5 \\ 1 & 2 \end{bmatrix}$$

$$M2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Input

lab5 task 4.py > ...

```python
import numpy as np

M1 = np.array([[5, 5],
               [1, 2]])
M2 = np.array([[0, 1],
               [1, 0]])
print("Matrix M1:")
print(M1)
print("\nMatrix M2:")
print(M2)
result = np.dot(M1, M2)
print("\nResult of M1 x M2:")
print(result)
transpose_result = result.T
print("\nTranspose of the resultant matrix:")
print(transpose_result)
```

Output

```
task 4.py"
Matrix M1:
[[5 5]
 [1 2]]

Matrix M2:
[[0 1]
 [1 0]]

Result of M1 x M2:
[[5 5]
 [2 1]]

Transpose of the resultant matrix:
[[5 2]
 [5 1]]
PS S:\AI Lab>
Result of M1 x M2:
[[5 5]
 [2 1]]

Transpose of the resultant matrix:
[[5 2]
 [5 1]]
Result of M1 x M2:
[[5 5]
```

```
Result of M1 x M2:
[[5 5]
 [2 1]]

Transpose of the resultant matrix:
Result of M1 x M2:
[[5 5]
 [2 1]]

Result of M1 x M2:
[[5 5]
Result of M1 x M2:
[[5 5]
Result of M1 x M2:
Result of M1 x M2:
[[5 5]
 [2 1]]

Transpose of the resultant matrix:
[[5 2]
 [5 1]]
PS S:\AI Lab>
```

Task 5. Find inverse of matrix

[[6, 1, 1, 3],

[4, -2, 5, 1],

[2, 8, 7, 6],

[3, 1, 9, 7]]

Input

```python
lab5 task 5.py > ...
import numpy as np
M = np.array([[6, 1, 1, 3],
              [4, -2, 5, 1],
              [2, 8, 7, 6],
              [3, 1, 9, 7]])

print("Original Matrix:")
print(M)
M_inv = np.linalg.inv(M)
print("\nInverse of the matrix:")
print(M_inv)
```

Output

```
task 5.py"
Original Matrix:
[[ 6  1  1  3]
 [ 4 -2  5  1]
 [ 2  8  7  6]
 [ 3  1  9  7]]

Inverse of the matrix:
[[ 0.13368984  0.10695187  0.02139037 -0.09090909]
 [-0.00229183  0.02673797  0.14820474 -0.12987013]
 [-0.12987013  0.18181818  0.06493506 -0.02597403]
 [ 0.11000764 -0.28342246 -0.11382735  0.23376623]]
PS S:\AI Lab>
```

Conclusion

This lab demonstrated the effective use of Python and NumPy for array manipulation and linear algebra operations. We successfully flattened arrays, identified maximum and minimum values, created structured matrices, and converted data types for precision. Matrix multiplication, transposition, and inversion highlighted the importance of these operations in solving complex problems. Overall, the tasks reinforced how computational tools provide accuracy, efficiency, and reliability in mathematical problem-solving.